Intel Galileo: A Comparison with Arduino Uno

This is the first part of a multi-part series on the Intel Galileo board, comparing it with the Arduino board. Hobbyists will find the articles in this series of great interest.

Over the years, hardware prototype development boards have grown exponentially and that has significantly reduced the amount of effort required for hardware development. Today, this is no longer the forte of the electronics engineer. Arduino, an open source easy-to-use platform, has played a pivotal role in this development. Anyone with basic electronics knowledge can easily get started with Arduino and can develop one's own projects. After observing Arduino's growth, Intel decided to launch custom hardware prototype boards based on its own technology. But Arduino and related hardware had become so popular that Intel's hardware boards didn't get adopted on the scale the company had expected. Subsequently, Intel launched the Galileo board, with the intention of becoming a part of the Arduino ecosystem and maker community.

So what is Intel Galileo?

Intel Galileo is the first Arduino-certified development board based on Intel technology. It's the first board based on Intel architecture designed to be hardware and software pin-compatible with Arduino shields designed for the Uno R3. It is capable of acting as the brain in a robot, controlling haunted house special effects, uploading sensor data to the Internet, and much more.

And it comprises the following:

1. **Processor:** Intel's Quark SoC X1000 application processor, designed for small-sized, low-power applications.
2. **RAM:** 512KB of RAM built into the processor and an additional 256MB on the chips.
3. **Flash memory:** Unlike with RAM, any data stored here is saved even after the board is shut down and the power is disconnected. It can hold 8MB of data, most of which is taken up by Galileo's operating system.
4. **MicroSD card slot:** If you need more space to run your program, you can use this slot, with up to 32GB. For advanced operations like running a Wi-Fi router, you have to use this slot because 8MB of Flash memory can't load drivers for such operations.
5. **Arduino expansion pins:** These are similar to Arduino Uno pins. You can use them as normal Arduino pins and connect different Arduino shields as well.